

RES.6:012 Introduction to Probability - Lecture 19/20
Textbook Problem Sets - Sections 5.4 & 9.1

• Problem Set 5.4

8) 100 rounds $\rightarrow \{1, 36\} \rightarrow$ odd/even

$$P(S_n \geq 55) = 1 - P(S_n \leq 55)$$

De Moivre-Laplace Binomial approximation:

$$\begin{aligned} P(54.5 \leq S_n \leq 55.5) &= \Phi\left(\frac{55.5 - 100(0.5)}{\sqrt{100(0.5)(0.5)}}\right) - \Phi\left(\frac{54.5 - 100(0.5)}{\sqrt{100(0.5)^2}}\right) \\ &= \Phi(1.1) - \Phi(0.9) \\ &= (0.3643) - (0.3159) \\ &= 0.0484 \end{aligned}$$

$$\begin{aligned} 1 - P(S_n \leq 55) &= 1 - P(54.5 \leq S_n \leq 55.5) \\ &= 1 - 0.0484 = 0.9516 \end{aligned}$$

$$n = 100$$

$$p = 0.5$$

$$E[X] = np = 100(0.5) = 50 \quad (\text{Binomial})$$

$$\text{var}(S_n) = \sigma_s = \sqrt{np(1-p)} = \sqrt{100(0.5)^2} = 5$$

$$\begin{aligned} P(S_n > 55) &= P\left(\frac{S - 50}{5} > \frac{55 - 50}{5}\right) \approx 1 - \Phi(1) \\ &= 1 - 0.8413 \end{aligned}$$

$$= 1 - P(S \leq 55.5)$$

$$= P\left(\frac{S - 50}{5} > \frac{55.5 - 50}{5}\right)$$

$$\approx 1 - \Phi(1.1) = 1 - 0.8643 = 0.1357$$

9) Crash $\rightarrow$ 5% $n=50$ ~~$\mu=0.05$~~ $p=1-0.05=0.95$

a.) ~~$P(S_n \geq 45)$~~ $E[S_n] = n\mu = 50(0.95) = 47.5$

$$\begin{aligned} P(S_n \geq 45) &= 1 - P(S_n \leq 44) \\ &= 1 - P(S_n \leq 44.5) \\ &= 1 - \Phi\left(\frac{44.5 - 47.5}{\sqrt{50(0.95)(0.05)}}\right) \\ &= 1 - \Phi(-1.94665) \\ &= 1 - 0.0256 \\ &= \underline{0.9744} \end{aligned}$$

10) $\mu=5$ $\text{var}=9$ $E[X_{100}] = n\mu = 100(5) = 500$

a) $P(X_{100} \leq 440) = P(X_{100} \leq 440) = \Phi\left(\frac{440 - 500}{\sqrt{3 \cdot 100}}\right)$

$$= \Phi(-2.0) = 0.0228$$

$\rightarrow P(X_{100} < 440) = P(X_{100} \leq 439.5)$

$\therefore \Phi(2.02) = 0.0217$

b) $P(X_1 + \dots + X_n \geq 200 + 5n) \leq 0.05$

$$\begin{aligned} P(S_n \geq 200 + 5n) &= 1 - P(S_n < 200 + 5n) = 0.05 \\ &= 1 - \Phi\left(\frac{200}{3\sqrt{n}}\right) \\ &= \Phi\left(\frac{200}{3\sqrt{n}}\right) \end{aligned}$$

$$= P\left(\frac{S_n - 5n}{3\sqrt{n}} \geq \frac{200}{3\sqrt{n}}\right) \leq 0.05$$

$$= 1 - \Phi\left(\frac{200}{3\sqrt{n}}\right) = 0.05$$

$$= \Phi\left(\frac{200}{3\sqrt{n}}\right) = 0.95$$

$\therefore \frac{200}{3\sqrt{n}} \geq 1.65$
 $\sqrt{n} \geq 40.4 \quad n \geq 1632$

c) ~~P(X_N > 1000)~~

$$P(X_{220} + \dots + X_N \geq 1000) =$$

$$P(S_n > 1000) = 1 - P(S_n \leq 1000)$$

$$= 1 - \cancel{\Phi} \left(\frac{1000 - 220(5)}{3\sqrt{220}} \right)$$

$$= 1 - \Phi(-2.2473)$$

$$= 1 - 0.0122$$

$$= \underline{0.9878} \quad X$$

$$P(N \geq 220) = P(S_{\overset{X}{219}} \leq 1000)$$

$$= P\left(\frac{S_{219} - 5(219)}{3\sqrt{219}} \leq \frac{1000 - 5(219)}{3\sqrt{219}}\right)$$

$$= 1 - \Phi(2.14)$$

$$= 1 - 0.9838$$

$$= \underline{0.0162}$$